LAB EXPERIMENTS

01) Select any system (ATM system) and thoroughly examine its specification to identify and document the different software defects.

Feature to be tested:

1. Validity of ATM card
2. withdrawal amount
3. Authentication of user
4. Dispersing Cash
5. Amount Balance Verification
6. Limit per transaction
7. Daily limit
8. Receipt Printing

Bug ID	Bug Name
ATM 001	Invalid Card
ATM 002	Invalid Pin
ATM 003	Invalid Acct type
ATM 004	Insufficient Balance
ATM 005	Transaction limit.
ATM 006	Limit Exceeded
ATM 007	Receipt Not Printed
ATM 008	Pin change mismatch.
ATM 009	Invalid Amount Information.

Bug ID : ATM001

Description : Invalid Card

steps : Insert an invalid or expired ATM card

Expected Result: ATM should display "Invalid Card" message

Actual Result : ATM displays welcome screen.

Status : Pass / Fail.

Bug ID : ATM 002

Bug Desc : Invalid Pin

steps : Insert Valid ATM card
Enter Incorrect Pin
Submit the PIN

Expected Result : " Invalid Pin" & deny Access

Actual Result : ATM displays menu screen.

Status : Pass / Fail

Bug ID : ATM 003

Bug Desc : Invalid Account type

Steps : Insert a valid ATM card and enter correct Pin.
Select the withdrawal option.
Select an unsupported / invalid account type.

Expected Result : ATM should reject invalid account.

Actual Result : ATM displays amount-entry scree.

Status : Pass / Fail.

Bug ID : ATM 004

Bug Desc : Insufficient Balance

Steps : Insert a valid ATM card.
Enter correct Pin
Select correct Account type.
Enter a larger amount
Confirm the transaction.

Expected Result : ATM should display "Insufficient Balance" message.

Actual Result : ATM deducts the amount if defect occurs.

Status : Pass / Fail.

Bug ID : ATM 005

Bug Desc : Withdrawal limit per transaction.

Steps : Insert a valid ATM card
Enter correct Pin
Select withdrawal option
Enter amount beyond the limit.
Confirm the transaction.

Expected Result : ATM should display "error message.

Actual Result : ATM goes back to home screen.

Status : Pass / Fail.

Q2) Arms portal - positive test scenario & test cases.

Positive test scenario :

Test Case ID : TC01

Test case : Login with valid credentials

Condition : User must have registered Account

Steps : Open arms portal
Enter valid username
Enter valid password
Click login Button.

Test data : Valid username and password.

Expected Result : User should log in successfully.

Status : Pass / Fail.

Test Case ID : TC02

Test Case : View student Profile

Condition : student must be logged in

Steps : Log in to portal
Open profile section
View student details.

Test data : Registered student Account

Expected Result : Correct details displayed.

Status : Pass / Fail.

Test Case ID : TC03

Test case : View attendence details.

Precondn : Students must be logged in and attendance record must be available.

Steps : Log in to the portal
Open attendance section
Select any sem or subject.

Test data : Valid student account with attendance records.

Status : Pass / Fail.

Test Case ID : TC04

Test Case : View Internal marks

Precondn : Internal marks must be published.

Steps : Log in to portal
Open marks section
Select required subject or semester.

Test data : Student account with published marks.

Expected Results : Correct internal marks displayed.

Status : Pass / Fail.

Test Case ID : TC05

Test Case : View Class Timetable

Precondn : Timetable must be available.

Steps : Log in to Portal

Open timetable section.
Select current semester.

Test Data : Valid Student Account

Expected Result : Correct class schedule, subject timings.

Status : Pass / Fail.

Q3) Arms Portal - Negative Test Scenarios & Test Cases.

Negative Test Cases :

Test Case ID : NT01

Test Case : Login with incorrect password.

Precondn : User must have registered Account.

Steps : Open ARMS portal
Enter valid username
Enter incorrect password.
Click login.

Test Data : Valid username & Incorrect Password.

Expected Result : Login should Fail.

Status : Pass / Fail.

Test Case ID : NT02

Test case : Login with unregistered username.

Precondn : ARMS portal login page must be available.

Steps : Open the portal
Enter an unregistered username
Enter a password.
Click login.

Test data: Unregistered username.

Expected Result: System should reject login attempt.

Status: Pass / Fail.

Testcase ID: NT03

Testcase: Submit login form with empty fields.

Precondn: Login page must be displayed.

Steps: Open the portal

Leave username, password fields Empty.

Click login.

Test data: Empty username, password.

Expected Result: System should display required field validation and login.

Status: Pass / Fail.

Test Case ID: NT04

Test case: Enter invalid register number.

Precondn: Submit registration (or) search form.

Steps: Open student details form

Enter invalid register number.

Click submit / search.

Test data: @@@#-#-##

Expected Result: System should reject invalid register number, display message.

Status: Pass / Fail.

Q4) E-Commerce Application - positive test scenario & test cases.

Positive Test Cases:

Test Case ID: ETC01

Test Case: Register with valid details.

Precondn: Registration page must be available.

Steps: Open E-commerce application

Click Register

Enter valid name, email & password.

Click submit.

Test data: Valid Name, unique-email, valid password.

Expected Result: Account should be created successfully.

Status: Pass / Fail.

Test Case: ETC02.

Test Case: login with valid credentials.

Precondn: User must have a registered account.

Steps: Open login page

Enter registered email & correct password.

click login.

Test Data: Valid email & password.

Expected Resulted: User should log successfully,

status: Pass / Fail.

Test Case ID: ETC03

Test Case: Search for existing product.

Precondn: Products must be available in location.

Steps to Execute: Open Application
Enter product name in search bar.
Click search.

Test data: Laptop.

Expected Result: Relevant Products should be displayed.

Status: Pass / Fail.

Test Case ID: ETC04

Test Case: View Product details.

Precondn: Product must exist in catalogue

Steps to Execute: Search for product
Select Required Product
Open details page

Test data: An existing product.

Expected Result: Correct product name, price must be displayed.

Status: Pass / Fail.

Test Case ID: ETC05

Test Case: Add product to existing shopping card.

Precondn: Product must be in stock.

Steps : Open product Page
Click Add to cart
Open shopping cart.

Test data : Available product, Quantity 1.

Expected Result : Select product should appear in cart.

Status : Pass/Fail.

05) E-commerce Application - Negative Test Scenario & Test cases.

Negative Test Scenario :

Test Case ID : NTC01

Test case : Register with invalid email address.

Precondn : Registration page must be available.

Steps : Open registration page
Enter name & invalid email address.
Enter valid Password
Click Submit.

Test data : Email abc123

Expected Result : Application should reject email and display message.

Status : Pass/Fail.

Test Case ID : NTC02

Test case : Login with incorrect password.

Precondn : User must have registered account.

Steps : Open login page
Enter registered email.
Enter incorrect password.
Click login.

Test Data : Registered email & incorrect password.

Expected Result : Login should fail, appropriate error message should be displayed.

Status : Pass / Fail.

Test Case ID : NTC03

Test Case : Submit Registration with empty fields.

Precondn : Registration page must be displayed.

Steps : Open Registration page.
Leave Required fields empty.
Click submit.

Tests data : Empty name, email & password.

Expected Result : Application should display required field messages.

Status : Pass / Fail.